

Solution Requirements

Date	14 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The Personalized Networking Assistant is an AI-powered web application with a secure Node/Express backend, Google Gemini generation, Wikipedia factual verification, persistent SQLite storage, subscription configuration, conversation history, feedback collection, and an analytical SaaS dashboard. The requirements below reflect the implemented project specification.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Account System	Secure registration, login, and detailed professional profile customization including role, interests, and career goals.	High
2	Subscription & Pricing Configuration	Support basic, executive, and enterprise modes with custom price or budget inputs for executive and enterprise plans.	High
3	Wikipedia Factual Verification	Search the Wikipedia API for event-specific domains and return factual summaries before AI generation.	High
4	Generative AI Icebreaker Engine	Use Google GenAI SDK with gemini-3.5-flash to generate structured talking points such as warm openers, elevator pitches, and LinkedIn follow-ups.	High

Step 1: Functional Requirements (FR) — Continued

S.No	Requirement Category	Requirement Description	Priority
5	Connection Logging & History	View persistent history of previous event generations, with options to delete logs and clear stored database rows.	High
6	Structured Feedback Loops	Rate generated starters from 1–5 stars, record likes or dislikes, and submit qualitative text feedback.	Medium
7	Analytical SaaS Dashboard	Render real-time graphs for history volume, average satisfaction ratings, and like-to-dislike ratios.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Security	All API keys, including GEMINI_API_KEY, must be managed server-side through process.env. Client-side key exposure is prohibited.	No exposed client API keys
2	Performance	Local database queries must execute quickly, and AI generation should complete promptly under normal network conditions.	DB queries under 20 ms; AI generation within 3 seconds
3	Responsive Design	The interface must render fluidly on standard monitors, tablets, and smartphone screen heights using Tailwind breakpoints.	Responsive across supported screen sizes

Step 2: Non-Functional Requirements (NFR) — Continued

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
4	Reliability	The application must gracefully transition to rich local default mock data if Gemini or Wikipedia encounters a network disruption.	Core interface remains usable during external API failure

Implementation Alignment

- Backend and API security are handled through the Node/Express server.
- Google Gemini (gemini-3.5-flash) provides structured AI-generated networking content.
- Wikipedia API provides factual grounding before generation.
- SQLite provides persistent storage for users, history, settings, and feedback.
- The frontend uses responsive Tailwind breakpoints and presents analytics through the SaaS dashboard.